process but, conditional on the filtration generated by λ itself, the process N whose first jump represents default remains a time-inhomogeneous Poisson process with intensity $\gamma = \lambda$.

Our assumptions on the short-rate process r and on the intensity dynamics come from the earlier part of the book, and this is where we actually use the interest rate models technology also for the default event, almost for free.

In particular, the model we consider is the CIR++ model introduced in Section 3.9. Although we will recall here some of the features of this model, we advise the reader who is not familiar with the CIR (Section 3.2.3) and CIR++ (Section 3.9) models to have a look at the earlier sections and related references to other parts of the book before proceeding further. We give a summary anyway, with a slightly generalized notation, so as to be able to distinguish the CIR++ component for interest rates from the CIR++ component for stochastic intensities. The model we introduce is called "Shifted Square Root Diffusion" (SSRD) model, and is based on Brigo and Alfonsi (2003, 2005).

22.7.1 A two-factor shifted square-root diffusion model for intensity and interest rates (Brigo and Alfonsi (2003))

We begin by illustrating the interest-rate part of the model.

CIR++ short-rate model (Brigo and Mercurio (2001a))

We write the short-rate r_t as the sum of a deterministic function φ and of a Markovian process x_t^α:

$$r_t = x_t^\alpha + \varphi(t;\alpha) \ , \quad t \geq 0, \tag{22.5}$$

where φ depends on the parameter vector α (which includes x_0^α) and is integrable on closed intervals. Notice that x_0^α is indeed one more parameter at our disposal: we are free to select its value as long as

$$\varphi(0;\alpha) = r_0 - x_0 \ .$$

We take as reference model for x the Cox-Ingersoll-Ross (1985) process:

$$dx_t^\alpha = k(\theta - x_t^\alpha)dt + \sigma\sqrt{x_t^\alpha}dW_t,$$

where the parameter vector is $\alpha = (k, \theta, \sigma, x_0^\alpha)$, with k, θ, σ, x_0^α positive deterministic constants. As usual, W is a standard Brownian motion process under the risk neutral measure, representing the stochastic shock in our dynamics. The condition

$$2k\theta > \sigma^2$$

ensures that the origin is inaccessible to the reference model, so that the process x^α remains positive. As is well known, this process x^α features a

noncentral *chi-square* distribution, and yields an affine term-structure of interest rates. Accordingly, analytical formulas for prices of zero-coupon bond options, caps and floors, and, through Jamshidian's decomposition, coupon-bearing bond options and swaptions, can be derived (see also Section 3.2.3). We can therefore consider the CIR++ model, consisting of our extension (22.5), and calculate the analytical formulas implied by such a model, by simply adapting the analogous explicit expressions for the reference CIR model as given in Cox et al. (1985). Denote by f instantaneous forward rates, i.e. $f(t,T) = -\partial \ln P(t,T)/\partial T$.

The initial market zero-coupon interest-rate curve $T \mapsto P^{\text{mkt}}(0,T)$ is automatically calibrated by our model if we set $\varphi(t;\alpha) = \varphi^{\text{CIR}}(t;\alpha)$ where

$$\begin{aligned} \varphi^{\text{CIR}}(t;\alpha) &= f^{\text{mkt}}(0,t) - f^{\text{CIR}}(0,t;\alpha), \\ f^{\text{CIR}}(0,t;\alpha) &= 2k\theta \frac{(\exp\{th\}-1)}{2h+(k+h)(\exp\{th\}-1)} \\ &\quad + x_0 \frac{4h^2 \exp\{th\}}{[2h+(k+h)(\exp\{th\}-1)]^2} \end{aligned} \tag{22.6}$$

with

$$h = \sqrt{k^2 + 2\sigma^2}.$$

For restrictions on the α's that keep r positive see Section 3.9.3.

The price at time t of a zero-coupon bond maturing at time T is

$$\begin{aligned} P(t,T) = \frac{P^{\text{mkt}}(0,T)A(0,t;\alpha)\exp\{-B(0,t;\alpha)x_0\}}{P^{\text{mkt}}(0,t)A(0,T;\alpha)\exp\{-B(0,T;\alpha)x_0\}} \cdot \\ \cdot P^{\text{CIR}}(t,T,r_t - \varphi^{\text{CIR}}(t;\alpha);\alpha) \end{aligned} \tag{22.7}$$

where

$$P^{\text{CIR}}(t,T,x_t;\alpha) = \mathbb{E}_t(e^{-\int_t^T x^\alpha(u)du}) = A(t,T;\alpha)\exp\{-B(t,T;\alpha)x_t\}$$

is the bond price formula for the basic CIR model given in Section 3.2.3; from this bond formula continuously compound spot rates $R(t,T)$ (still affine in r_t), the spot LIBOR rates $L(t,T)$, forward LIBOR rates $F(t,T,S)$ and all other kind of rates can be easily computed as explicit functions of r_t. We omit the argument α when clear from the context.

The cap option price formula for the CIR++ model can be derived easily in closed form from the corresponding formula for the basic CIR model. This formula is a function of the parameters α. In our application we will calibrate the parameters α to cap prices, by inverting the analytical CIR++ formula, so that our interest rate model is calibrated to the initial zero coupon curve through φ and to the cap market through α. See also the calibration example in Section 3.14.

After summarizing the interest-rate part of the model, we now move to the intensity part of the model.

CIR++ intensity model (Brigo and Alfonsi (2003, 2005))

For the intensity model we adopt a similar approach, in that we set

$$\lambda_t = y_t^\beta + \psi(t;\beta) \ , \quad t \geq 0, \tag{22.8}$$

where ψ is a deterministic function, depending on the parameter vector β (which includes y_0^β), that is integrable on closed intervals. As before, y_0^β is indeed one more parameter at our disposal: We are free to select its value as long as

$$\psi(0;\beta) = \lambda_0 - y_0 \ .$$

We take y again of the form:

$$dy_t^\beta = \kappa(\mu - y_t^\beta)dt + \nu\sqrt{y_t^\beta}dZ_t,$$

where the parameter vector is $\beta = (\kappa, \mu, \nu, y_0^\beta)$, with κ, μ, ν, y_0^β positive deterministic constants. As usual, Z is a standard Brownian motion process under the risk neutral measure, representing the stochastic shock in our dynamics. Again we assume the origin to be inaccessible, i.e.

$$2\kappa\mu > \nu^2.$$

For restrictions on the β's that keep λ positive, as is required in intensity models, see again the suggestions given in Section 3.9.3.

We will often use the integrated quantities

$$\Lambda(t) = \int_0^t \lambda_s ds, \quad Y^\beta(t) = \int_0^t y_s^\beta ds, \quad \text{and} \ \ \Psi(t,\beta) = \int_0^t \psi(s,\beta)ds.$$

We take the short interest-rate and the intensity processes to be correlated, by assuming the driving Brownian motions W and Z to be instantaneously correlated according to

$$dW_t \ dZ_t = \rho \ dt.$$

This way to model the intensity and the short interest rate can be viewed as a generalization of a particular case of the Lando's (1998) approach, and can also be seen as a generalization of a particular case of the Duffie and Singleton (1997, 1999) square-root diffusion model (see for example Bielecki and Rutkowski (2001), pp 253-258). In both cases we add a non homogeneous term to recover exactly fundamental market data in the spirit of our earlier deterministic shift extension of Section 3.8 (see also Brigo and Mercurio (2001a)).

22.7.2 Calibrating the joint stochastic model to CDS: Separability

With the above choice for λ, in the credit derivatives world we have formulas that are analogous to the ones for interest-rate derivatives products. Consider for example the risk-neutral survival probability. We have easily, as we have seen several times, that

$$\mathbb{Q}(\tau > t) = \mathbb{E}[1_{\{\tau>t\}}] = \mathbb{E}[\mathbb{E}(1_{\{\tau>t\}}|\mathcal{F}^{\lambda})] =$$
$$= \mathbb{E}[\mathbb{E}(1_{\{\Lambda(\tau)>\Lambda(t)\}}|\mathcal{F}^{\lambda})] = \mathbb{E}e^{-\Lambda(t)} = \mathbb{E}(e^{-\int_0^t \lambda(u)du}),$$

since, conditional on λ, $\Lambda(\tau)$ is an exponential random variable ξ with parameter one. Notice that, if λ were a short-rate process, the last expectation of the "stochastic discount factor" would simply be the zero-coupon bond price in our interest-rate model. In other terms, notice the analogy we already pointed out:

$$P(0,t) = \mathbb{E}(e^{-\int_0^t r(u)du}), \quad \mathbb{Q}(\tau > t) = \mathbb{E}[1_{\{\tau>t\}}] = \mathbb{E}(e^{-\int_0^t \lambda(u)du}).$$

So we see that survival probabilities for the λ model are the analogous of zero-coupon bond prices P in the r model. Thus if we choose for λ a CIR++ process, survival probabilities will be given by the CIR++ model bond price formula.

In particular, we can express credit default swaps data through the implied hazard function Γ according to the method described in Section 21.3.5. The related implied Γ, stripped from CDS data, will be denoted by $\Gamma = \Gamma^{\text{mkt}}$, while the associated hazard rate will be denoted by $\gamma = \gamma^{\text{mkt}}$.

Now assume $\rho = 0$. This implies that the intensity λ and the cumulated intensity Λ are independent of the short rate r, and of interest rates in general. Since $\tau = \Lambda^{-1}(\xi)$ and ξ is also independent of interest rates, in this $\rho = 0$ case the default time τ and interest rate quantities $r, D(s,t), \ldots$ are independent. It follows that everything we stated on CDS valuation in Section 21.3.5 holds, and in particular the model independent formulas (21.22) and (21.23) hold. In other terms, a CDS valuation (say at time 0 and seen from the protection seller) when $\rho = 0$ becomes model independent and is given by the formula

$$\text{CDS}_{a,b}(0,R;P(0,\cdot),\mathbb{Q}(\tau>\cdot)) = R\Big[-\int_{T_a}^{T_b} P(0,t)(t-T_{\beta(t)-1})d_t\mathbb{Q}(\tau \geq t)$$
$$+\sum_{i=a+1}^{b} P(0,T_i)\alpha_i\mathbb{Q}(\tau \geq T_i)\Big] + \text{L}_{\text{GD}}\int_{T_a}^{T_b} P(0,t)\ d_t\mathbb{Q}(\tau \geq t) \qquad (22.9)$$

This formula is indeed model independent given the initial zero coupon curve (bonds) at time 0 observed in the market (i.e. $P(0,\cdot)$) and given the survival probabilities $\mathbb{Q}(\tau \geq \cdot)$ at time 0. However, this is also the formula for the SSRD model corresponding to the case $\rho = 0$. This means that if we strip survival probabilities from CDS in a model independent way, to calibrate the

market CDS quotes we just need to make sure that the survival probabilities we strip from CDS are correctly reproduced by the SSRD model. Since the survival probabilities in the SSRD model are given by

$$\mathbb{Q}(\tau > t)_{model} = \mathbb{E}(e^{-\Lambda(t)}) = \mathbb{E}\exp\left(-\Psi(t,\beta) - Y^{\beta}(t)\right) \tag{22.10}$$

we just need to make sure

$$\mathbb{E}\exp\left(-\Psi(t,\beta) - Y^{\beta}(t)\right) = \mathbb{Q}(\tau > t)_{market}.$$

If we agree to express survival probabilities through implied hazard rates and functions, so that $\mathbb{Q}(\tau > t)_{market} = e^{-\Gamma^{\mathrm{mkt}}(t)}$, then Equation (22.10) reads

$$\mathbb{E}\exp\left(-\Psi(t,\beta) - Y^{\beta}(t)\right) = e^{-\Gamma^{\mathrm{mkt}}(t)} \tag{22.11}$$

from which

$$\Psi(t,\beta) = \Gamma^{\mathrm{mkt}}(t) + \ln(\mathbb{E}(e^{-Y^{\beta}(t)})) = \Gamma^{\mathrm{mkt}}(t) + \ln(P^{\mathrm{CIR}}(0,t,y_0;\beta)), \tag{22.12}$$

where we choose the parameters β in order to have a positive function ψ (i.e. an increasing Ψ). Thus, if ψ is selected according to this last formula, as we will assume from now on, the model is calibrated to the market implied hazard function Γ^{mkt}, i.e. to CDS data.

The whole chain of reasoning relies on the possibility to disentangle survival probabilities from discount factors, in a way that leads to the model independent formula (22.9) for CDS prices. This allows us to calibrate directly the survival probabilities of the model to the survival probabilities from the market. The condition of independence between τ and r is fundamental for this disentangling result to hold, and is satisfied only when $\rho = 0$.

Indeed, if $\rho \neq 0$ then Formula (22.9) no longer holds and to calibrate cds data we need to solve

$$\mathrm{CDS}^{\mathrm{SSRD}}_{0,b}(0, R^{\mathrm{MKT}}_{0,b}(0); \alpha, \beta, \varphi(\cdot;\alpha), \psi(\cdot,\beta), \rho) = 0$$

for a set of canonical maturities T_b, typically ranging from one year to ten years, jointly with analogous equations for interest rate derivatives involving α and φ. This involves the joint parameters of the r and λ models and the correlation ρ to be calibrated all at the same time.

Let us now compute the CDS price in the SSRD model in general.

The $\mathrm{CDS}_{a,b}(t,R,\mathrm{L_{GD}})$ price in a stochastic intensity model is given by

$$\begin{aligned}\mathrm{CDS}_{a,b}(t,R,\mathrm{L_{GD}}) &= 1_{\{\tau>t\}}\mathbb{E}\Big\{D(t,\tau)(\tau - T_{\beta(\tau)-1})R\mathbf{1}_{\{T_a<\tau<T_b\}} \\ &\quad + \sum_{i=a+1}^{b} D(t,T_i)\alpha_i R\mathbf{1}_{\{\tau>T_i\}} - \mathbf{1}_{\{T_a<\tau<T_b\}}D(t,\tau)\ \mathrm{L_{GD}}|\mathcal{G}_t\Big\}.\end{aligned}$$

Let us expand the first term. We obtain

$$
\begin{aligned}
&1_{\{\tau>t\}}\mathbb{E}\Big\{D(t,\tau)(\tau-T_{\beta(\tau)-1})R1_{\{T_a<\tau<T_b\}}\Big|\mathcal{G}_t\Big\}\\
&=\frac{1_{\{\tau>t\}}}{\mathbb{Q}(\tau>t|\mathcal{F}_t)}\mathbb{E}\Big\{D(t,\tau)(\tau-T_{\beta(\tau)-1})R1_{\{T_a<\tau<T_b\}}\Big|\mathcal{F}_t\Big\}\\
&=\frac{1_{\{\tau>t\}}}{\exp(-\int_0^t\lambda_s ds)}\mathbb{E}\Big\{\int_t^\infty D(t,s)(s-T_{\beta(s)-1})R1_{\{T_a<s<T_b\}}1_{\{\tau\in[s,s+ds)\}}\Big|\mathcal{F}_t\Big\}=\\
&=\frac{1_{\{\tau>t\}}}{\exp(-\int_0^t\lambda_s ds)}\mathbb{E}\Big[\mathbb{E}\Big\{\int_{T_a}^{T_b} D(t,s)(s-T_{\beta(s)-1})R1_{\{\tau\in[s,s+ds)\}}\Big|\mathcal{F}_{T_b}\Big\}\Big|\mathcal{F}_t\Big]\\
&=\frac{1_{\{\tau>t\}}}{\exp(-\int_0^t\lambda_s ds)}\mathbb{E}\Big[\int_{T_a}^{T_b} D(t,s)(s-T_{\beta(s)-1})R\mathbb{E}\Big\{1_{\{\tau\in[s,s+ds)\}}\Big|\mathcal{F}_{T_b}\Big\}\Big|\mathcal{F}_t\Big]\\
&=\frac{1_{\{\tau>t\}}}{\exp(-\int_0^t\lambda_s ds)}\mathbb{E}\Big[\int_{T_a}^{T_b} D(t,s)(s-T_{\beta(s)-1})R\,\mathbb{Q}\Big\{\tau\in[s,s+ds)\Big|\mathcal{F}_{T_b}\Big\}\Big|\mathcal{F}_t\Big]\\
&=\frac{1_{\{\tau>t\}}}{\exp(-\int_0^t\lambda_s ds)}\mathbb{E}\Big[\int_{T_a}^{T_b} D(t,s)(s-T_{\beta(s)-1})R\exp\Big(-\int_0^s\lambda_u du\Big)\lambda_s ds\Big|\mathcal{F}_t\Big]\\
&=1_{\{\tau>t\}}\mathbb{E}\Big[\int_{T_a}^{T_b} D(t,s)(s-T_{\beta(s)-1})R\exp\Big(-\int_t^s\lambda_u du\Big)\lambda_s ds\Big|\mathcal{F}_t\Big]
\end{aligned}
$$

where we have used the filtration switching formula (22.3) in the first equality and equation (22.1) in the last two steps, since we are conditioning on $\mathcal{F}_{T_b}$, which makes intensity deterministic and thus formula (22.1) applicable, provided we substitute γ with λ (and t with s). By recalling that $D(t,s)=\exp(-\int_t^s r_u du)$ and carrying out analogous calculations for the remaining CDS terms, we can write eventually, under any stochastic intensity model:

$$
\begin{aligned}
\mathrm{CDS}_{a,b}(t,R,\mathrm{L_{GD}}) &= 1_{\{\tau>t\}}\Big\{R\sum_{i=a+1}^{b}\alpha_i\mathbb{E}\Big[\exp\Big(-\int_t^{T_i}(r_s+\lambda_s)ds\Big)\Big|\mathcal{F}_t\Big]\\
&+R\int_{T_a}^{T_b}\mathbb{E}\Big[\exp\Big(-\int_t^u(r_s+\lambda_s)ds\Big)\lambda_u\Big|\mathcal{F}_t\Big](u-T_{\beta(u)-1})du\\
&-\mathrm{L_{GD}}\int_{T_a}^{T_b}\mathbb{E}\Big[\exp\Big(-\int_t^u(r_s+\lambda_s)ds\Big)\lambda_u\Big|\mathcal{F}_t\Big]du\Big\} \qquad (22.13)
\end{aligned}
$$

When considering the above expected values in the SSRD model with $\rho\neq 0$, given the particular λ and r processes and the parameters in their dynamics, we obtain a formula $\mathrm{CDS}_{a,b}^{\mathrm{SSRD}}(0,R;\alpha,\beta,\varphi(\cdot;\alpha),\psi(\cdot,\beta),\rho)$ for CDS valuation in the SSRD model. However, this formula admits no closed form. One is then forced in principle to solve

$$\mathrm{CDS}_{0,b}^{\mathrm{SSRD}}(0, R_{0,b}^{\mathrm{MKT}}(0); \alpha, \beta, \varphi(\cdot;\alpha), \psi(\cdot,\beta), \rho) = 0 \tag{22.14}$$

numerically, together with the equations for interest rate derivatives calibration, in the unknowns $\alpha, \beta, \varphi(\cdot;\alpha), \psi(\cdot,\beta), \rho$. This may be a rather dramatic task. None of this problems is present when $\rho = 0$. Indeed, as a double-check we now compute the above general expression under the assumption that $\rho = 0$. The reader not interested in the double check may skip directly to the end of the double check part.

Begin double-check: Assuming $\rho = 0$ in the general CDS expression (22.13).

Assuming $\rho = 0$, the expectations appearing in the above expression can be computed as follows:

$$\begin{aligned}
&\mathbb{E}\left[\exp\left(-\int_t^{T_i}(r_s+\lambda_s)ds\right)|\mathcal{F}_t\right] = \\
&= \mathbb{E}\left[\exp\left(-\int_t^{T_i} r_s ds\right)\exp\left(-\int_t^{T_i}\lambda_s ds\right)\Big|\mathcal{F}_t\right] \\
&= \mathbb{E}\left[\exp\left(-\int_t^{T_i} r_s ds\right)\Big|\mathcal{F}_t\right]\mathbb{E}\left[\exp\left(-\int_t^{T_i}\lambda_s ds\right)\Big|\mathcal{F}_t\right] \\
&= \exp(\Psi(t,\beta)-\Psi(T_i,\beta))P^{\mathrm{CIR}}(t,T_i;y_t,\beta)\times \\
&\quad\times\exp(\Phi(t,\alpha)-\Phi(T_i,\alpha))P^{\mathrm{CIR}}(t,T_i;x_t,\alpha).
\end{aligned} \tag{22.15}$$

Further, again with $\rho = 0$, we may compute

$$\begin{aligned}
&\mathbb{E}\left[\exp\left(-\int_t^{u}(r_s+\lambda_s)ds\right)\lambda_u\Big|\mathcal{F}_t\right] = \\
&= \mathbb{E}\left[\exp\left(-\int_t^{u} r_s ds\right)\Big|\mathcal{F}_t\right]\mathbb{E}\left[\exp\left(-\int_t^{u}\lambda_s ds\right)\lambda_u\Big|\mathcal{F}_t\right] = \\
&= \mathbb{E}\left[\exp\left(-\int_t^{u} r_s ds\right)\Big|\mathcal{F}_t\right]\left(-\frac{d}{du}\mathbb{E}\left[\exp\left(-\int_t^{u}\lambda_s ds\right)\Big|\mathcal{F}_t\right]\right) = \\
&= -\exp(\Phi(t,\alpha)-\Phi(u,\alpha))P^{\mathrm{CIR}}(t,u;x_t,\alpha)\times \\
&\quad\times\frac{d}{du}\left[\exp(\Psi(t,\beta)-\Psi(u,\beta))P^{\mathrm{CIR}}(t,u;y_t,\beta)\right]
\end{aligned} \tag{22.16}$$

so that all terms are known analytically given x_t and y_t.

Now, if we are using the deterministic shifts φ and ψ coming from market data at time t given by P^{mkt} (or equivalently f^{mkt}) and Γ^{mkt}, according to the procedures outlined above (in particular see (22.6) and (22.12)), then by construction all the above terms give a CDS value that is consistent with

the model-independent CDS value where survival probabilities are re-cast in terms of hazard functions, according to formula (22.2) with $\Gamma = \Gamma^{\text{mkt}}$.

Indeed, with φ coming from (22.6) and ψ chosen via (22.12), bonds and survival probabilities are by construction consistent with model-independent market bonds and with model-independent survival probabilities, to which we associated the relevant market implied hazard functions, and expression (22.15) reduces to

$$P^{\text{mkt}}(t, T_i)\left(e^{-(\Gamma^{\text{mkt}}(T_i)-\Gamma^{\text{mkt}}(t))}\right)$$

while expression (22.16) reduces to

$$\begin{aligned} &-P^{\text{mkt}}(t,u)\frac{d}{du}\left(e^{-(\Gamma^{\text{mkt}}(u)-\Gamma^{\text{mkt}}(t))}\right) \\ &= P^{\text{mkt}}(t,u)\gamma^{\text{mkt}}(u)\left(e^{-(\Gamma^{\text{mkt}}(u)-\Gamma^{\text{mkt}}(t))}\right). \end{aligned}$$

If for simplicity we take the initial time $t = 0$ the two terms reduce to

$$P^{\text{mkt}}(0,u)\gamma^{\text{mkt}}(u)e^{-\Gamma^{\text{mkt}}(u)}, \quad P^{\text{mkt}}(0,T_i)e^{-\Gamma^{\text{mkt}}(T_i)}$$

and, when we substitute them in the CDS-price formula (22.13) valued at $t = 0$ we find the same expression as in Equation (22.2). For this to hold, i.e. for the SSRD model price to be consistent with the implied hazard rate price we needed to assume $\rho = 0$, leading to the factorizations given in formulas (22.15) and (22.16).

End of double-check.

To sum up, we have shown that in case $\rho = 0$,

i) Calibrating the model survival probabilities to the CDS-implied hazard function Γ^{mkt} by choosing ψ according to (22.12)

is equivalent to

ii) directly calibrate the (r, λ)-model by setting to zero the model CDS prices with the market quoted $R_{0,b}$'s in the premium legs.

More precisely, we have shown that if $\rho = 0$ and $\psi(\cdot;\beta)$ is selected according to (22.12), then the price ii) of the $\text{CDS}_{0,b}$ payoff under the stochastic intensity model λ is the same price i) obtained under hazard rate γ^{mkt} and is given by (22.2). So, as we knew already, when $\rho = 0$ the CDS price does not depend on the dynamics of (λ, r), and in particular it does not depend on $k, \theta, \sigma, \kappa, \nu$ and μ. We will verify this also numerically in Table 22.22: by amplifying intensity parameters through an increase of κ, ν and μ we do not substantially affect the CDS price in case $\rho = 0$.

However, if $\rho \neq 0$, the factorizations (22.15) and (22.16) on which the equivalence with an implied hazard function is based do not hold. As we

have seen in (22.14), the CDS price under the SSRD model becomes different and is in principle dependent on the dynamics; now the two procedures i) and ii) are not equivalent: The correct one would be to equate to zero the model CDS prices and solve Equation (22.14) (together with the interest derivatives equations), to obtain the model parameters, including in principle ρ.

This is annoying, since the attractive feature of the model is the separate and semi-automatic calibration of the interest-rate part to interest-rate data and of the intensity part to credit market data. Indeed, in the separable case the credit derivatives desk might ask for the α parameters and the $\varphi(\cdot;\alpha)$ curve to the interest-rate derivatives desk, and then proceed with finding β and $\psi(\cdot;\beta)$ from CDS data. This ensures also a consistency of the interest rate model that is used in credit derivatives evaluation with the interest rate model that is used for default-free derivatives. This separate automatic calibration no longer holds if we introduce ρ, since now the dynamics of interest rates is also affecting the CDS price.

However, we will see below in Table 22.22 that the impact of ρ is typically negligible on CDSs, even in case intensity parameters are increased by a factor from 3 to 5.

We can thus calibrate CDS data with $\rho = 0$, using the separate calibration procedure outlined above, and then set ρ to a desired value, based perhaps on historical estimation or on market views.

Once we have done this and calibrated CDS data through $\psi(\cdot,\beta)$, we are left with the parameters β, which can be used to calibrate further products, similarly to the way the α parameters of the r model are used to calibrate cap prices after calibration of the zero-coupon curve in the interest rate market. However, this will be interesting when option data on the credit derivatives market will become more liquid. Currently the bid-ask spreads for single name CDS options are large and suggest to consider these first quotes with caution. At the moment we content ourselves of calibrating only CDS's for the credit part. To help specifying β without further data we impose a constraint on the calibration of CDS's. We require the β's to be found to keep Ψ positive and increasing and to minimize $\int_0^T \psi(s,\beta)^2 ds$. This minimization amounts to contain the departure of λ from its time-homogeneous component y^β as much as possible. Indeed, if we take as criterion the integrated squared difference between "instantaneous forward rates" γ^{mkt} in the market and $f^{\text{CIR}}(\cdot;\beta)$ in our homogeneous CIR model with β parameters, constraining these differences to be positive at all points, the related minimization gives us the time-homogeneous CIR model β that is closest to market data under the given constraints.

We calibrated the same CDS data used to obtain the implied hazard function Γ given in Figure 22.16 up to a ten years maturity.

In Figure 22.17 we give the related piecewise linear hazard rate $\gamma^{\text{mkt}}(t)$ obtained by calibrating the 1y, 3y, 5y, 7y and 10y CDS's on Merrill-Lynch on October 2002. In Figure 22.16 the related risk-neutral default probabilities